COMPUTER-ALGEBRA VERIFICATION DETAILS

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This file accompanies *Laplacian algebras and subcalibrations for minimal hypercones*. References to sections, results, and numbered equations use the accompanying manuscript's numbering; equations prefixed by S are local to these notes. The programs and execution records are available at

https://github.com/HongbinCui/laplacian-subcalibrations.

The paper's Appendix G gives the concise overview.

This supplement records the algebraic identities checked by the ancillary programs. Symbolic identities and finite consistency tests are distinguished below. The geometric comparison, singular-set, and spectral-domain arguments, and the universal generator bound, are proved in the paper.

Running the programs. Download and fully extract the repository before opening a notebook. All paths below are relative to its `anc` directory. In a fresh Mathematica 12 kernel, open `START_HERE_Verification_v5.nb` and choose `Evaluation > Evaluate Notebook`. This runs the thirteen retained units, comprising 550 assertions. Each unit must pass its checks and export its text and CSV records to `mathematica/chapters/verification_results_v6` to return `True`. Individual units can be run from their notebooks in `mathematica/chapters`; the additional units run separately. The program filenames, context names, and output directories containing `v5`, `v6`, or dates are stable identifiers; they do not track manuscript versions or current chapter numbers. The correspondence is given in S.11; detailed commands and the inventory of native Mathematica and Python execution records are in `README.txt` and `Verification_Records.md`.

Execution status, 2 October 2026. All 660 Wolfram Language assertions have passed in the author's native Mathematica 12.0.0 runs on Windows 64-bit: 550 in the retained suite, 31 in the general cubic unit, 28 in the exceptional cubic unit, 38 in the cubic-update unit, and 13 in the pullback unit. The supplied evaluated notebooks and result tables preserve these runs. The three supplemental summaries record matching source hashes, successful evaluation, and no evaluation messages; their notebooks also confirm successful result exports. The two Python-only units retain passing records for 65 and 62 assertions. Thus the inventory contains 787 program assertions, including the stated repetitions; this update does not claim new Python executions.

For generators $y = (y_1, \dots, y_s)$, a symmetric matrix M, and a vector b, the program uses the differential operators

$$\begin{aligned} \Gamma(p,q) &= \sum_{i,j} M_{ij}\partial_i p\,\partial_j q, \\ \Delta p &= \sum_i b_i \partial_i p + \sum_{i,j} M_{ij}\partial_i\partial_j p, \\ \mathcal{L}(p) &= \Gamma(p,p)\Delta p - \tfrac{1}{2}\Gamma\big(p,\Gamma(p,p)\big). \end{aligned} \tag{S.1}$$

When $M_{ij} = \langle \nabla y_i, \nabla y_j\rangle$ and $b_i = \Delta y_i$, these are the ambient chain rules. Every identity check simplifies the difference of the two sides to zero. All numerical coefficients and test inputs are exact rational or algebraic numbers. Real powers are differentiated where their bases are positive.

Date: 3 October 2026, documentation update.

S.1. The gradient-power identity. Set $S = |\nabla F|^2$, $U = \Gamma(F, S)$, $V = \Gamma(S, S)$, $D = \Delta F$, and $B = \Delta S$. For a symbolic real exponent γ, let

$$H = S^\gamma F, \qquad \lambda_F = \frac{SD - U/2}{F}.$$

The calculation treats the further contractions $\Gamma(F, U), \Gamma(S, U), \Gamma(F, V), \Gamma(S, V)$ as independent symbols. Thus it does not require a particular polynomial closure. With

$$\begin{aligned}\mathcal{V}_\gamma &= S + \frac{2\gamma FU}{S} + \frac{\gamma^2 F^2 V}{S^2},\\ \mathcal{D}_\gamma &= D + \frac{2\gamma U}{S} + \frac{\gamma FB}{S} + \frac{\gamma(\gamma-1)FV}{S^2},\end{aligned}$$

the two residuals checked are

$$\Gamma(H, H) - S^{2\gamma}\mathcal{V}_\gamma = 0, \qquad S^{-3\gamma}\mathcal{L}(H) - F\mathcal{Q}_\gamma = 0, \tag{S.2}$$

where the Laplacian used in the second calculation is $\Delta H = S^\gamma \mathcal{D}_\gamma$, and

$$\begin{aligned}\mathcal{Q}_\gamma = \lambda_F + \gamma B &+ \frac{2\gamma UD - \gamma\Gamma(F, U) - (\gamma^2 + \frac{3}{2}\gamma)V}{S} + \frac{\gamma(\gamma+1)U^2}{S^2}\\ &+ \frac{\gamma^2 F}{S^2}\left(2UB + VD - \Gamma(S, U) - \tfrac{1}{2}\Gamma(F, V)\right)\\ &+ \frac{\gamma^3 F^2}{S^3}\left(VB - \tfrac{1}{2}\Gamma(S, V)\right).\end{aligned}$$

The symbolic substitution for λ_F is initially made where $F \neq 0$. Under the paper's hypothesis $\mathcal{L}(F) = F\lambda_F$ with smooth λ_F, the final formula also holds at regular zeros of F, where $S > 0$.

S.2. The radial master identity. Let H be homogeneous of degree m, and write $R = |x|^2$, $S = |\nabla H|^2$, $U = \Gamma(H, S)$, and $D = \Delta H$. Euler's identities supply the contractions

$$\Gamma(R, R) = 4R, \quad \Gamma(R, H) = 2mH, \quad \Gamma(R, S) = 4(m-1)S, \quad \Delta R = 2N.$$

For $G = R^\alpha H$ and $c = 4\alpha(m+\alpha)$, the program checks the following three identities with symbolic m, α, N:

$$\begin{aligned}R^{-2\alpha}\Gamma(G, G) &= S + cH^2/R,\\ R^{-\alpha}\Delta G &= D + 2\alpha(2m + N + 2\alpha - 2)H/R,\\ R^{-3\alpha}\mathcal{L}(G) &= SD - U/2 + \frac{cH^2 D}{R} + \frac{\big(2\alpha(N-1) - c\big)HS}{R}\\ &\qquad + \frac{c\big(m + 2\alpha(N-1)\big)H^3}{R^2}.\end{aligned}$$

These are symbolic identities for arbitrary degree and exponent; no list of sample exponents is used.

S.3. The normalized cubic closure. For the generators (R, F, S), the normalized closure is

$$M = \begin{pmatrix} 4R & 6F & 8S \\ 6F & S & 4RF \\ 8S & 4RF & 4RS + 12F^2 \end{pmatrix}, \qquad b = \big(2(6k-3), 0, 4kR\big).$$

Substitution of $G = SF$ in (S.1) gives

$$\begin{aligned}\Gamma(G, G) = V_k &:= S^3 + 12RSF^2 + 12F^4,\\ \Delta G &= (4k+8)RF,\end{aligned}$$

$$\mathcal{L}(G) = FQ_k,$$
$$Q_k = (4k-16)RS^3 + F^2S(48kR^2 - 126S) + 12(4k-6)RF^4.$$

The fourth check is the parameter-propagation identity

$$Q_k - Q_4 = 4(k-4)R\big(S^3 + 12RSF^2 + 12F^4\big).$$

This calculation isolates the dependence on k. The geometric restrictions on (R, F, S), and the sign at the initial parameter, are established in the paper.

S.4. Riedler's quartic potential. In the generator order (R, Q, F), use

$$M = \begin{pmatrix} 4R & 8Q & 8F \\ 8Q & 16RQ & 16RF \\ 8F & 16RF & 16RQ \end{pmatrix}, \qquad b = (2N, 16kR, 0).$$

On $R, Q > 0$, the potential is $G = RQ^{3/4}F$. Define

$$V = RQ^{1/2}\big(16R^2Q^2 + 33R^2F^2 + 32QF^2\big),$$
$$T = RQ^{1/4}\Big[16(12k-47)R^4Q^2 + 16(2N-41)R^2Q^3 + F^2\big((396k+66)R^4 + (384k+66N-681)R^2Q + (64N+160)Q^2\big)\Big].$$

The ambient residuals are

$$\Gamma(G,G) - V = 0, \qquad \mathcal{L}(G) - FT = 0,$$

together with

$$\det M = 1024R(R^2 - Q)(Q^2 - F^2).$$

Writing $a = RQ^{3/4}$, the boundary coefficient is checked as

$$\beta_G = \frac{RT|_{F=0}}{a^3\,16RQ} = 2N - 41 + (12k-47)\frac{R^2}{Q}.$$

The program also performs an independent spherical substitution. On $R = 1$, set $f = F$, $z = Q$, and $h = z^{3/4}f$. The spherical operator has matrix and drift

$$M_{\mathrm{sph}} = \begin{pmatrix} 16z - 16f^2 & 16f(1-z) \\ 16f(1-z) & 16z(1-z) \end{pmatrix}, \qquad b_{\mathrm{sph}} = \big(-4(N+2)f, 16k - 4(N+2)z\big).$$

Since G has degree nine, put

$$W = 81h^2 + \Gamma_{\mathrm{sph}}(h,h), \qquad A = W\big(\Delta_{\mathrm{sph}}h + 9(N-1)h\big) - \tfrac{1}{2}\Gamma_{\mathrm{sph}}(h, W).$$

The additional residuals are $W - V|_{(R,Q,F)=(1,z,f)} = 0$ and $A - (FT)|_{(R,Q,F)=(1,z,f)} = 0$.

S.5. Isoparametric certificates. For a polynomial $h = h(z)$, define

$$A(z) = g^2(1 - (z+t)^2), \qquad B(z) = -g^2t - g(g+d)z,$$
$$E(z) = \ell^2h^2 + A(h')^2, \qquad \mathcal{N}_\ell[h] = \big(Ah'' + Bh' + (d+1)\ell h\big)E - \tfrac{1}{2}Ah'E'.$$

The realizable interval is $[-1-t, 1-t]$. With $\mu = (d+1)\ell - \ell^2 - gd$ and $\nu = (d+1)\ell - gd$, the symbolic radial check is

$$\mathcal{N}_\ell[z] = z\left[g^2(1-t^2)\mu + g^2t(\ell^2 - 2\nu)z + (\ell^2 - g^2)\nu z^2\right].$$

For the first three corrections $h = z(1 + cz)$, the exact factorizations are $z\mathcal{N}_\ell[h] = z^4K(z)$, with

(g, d, t, ℓ)	c	$K(z)$
$(2, 6, 1/3, 4)$	$3/10$	$\frac{648}{125}(11z + 30)$
$(4, 14, 1/7, 8)$	$7/54$	$-\frac{64}{19683}(2548z^2 - 132237z - 910926)$
$(4, 14, 3/7, 8)$	$7/15$	$-\frac{128}{1125}(588z^2 - 8015z - 16550)$

Their positivity is checked using the exact Bernstein conversion (B.2) and its lower bound $P \geq \min_i b_i$. Using the degree of K in each row gives

(g, t)	$[a, b]$	$\min_i b_i(K)$
$(2, 1/3)$	$[-4/3, 2/3]$	$9936/125$
$(4, 1/7)$	$[-8/7, 6/7]$	$48414080/19683$
$(4, 3/7)$	$[-10/7, 4/7]$	$6656/15$

For the fourth correction, $(g, d, t, \ell) = (4, 16, 3/4, 10)$ and $h = z(1 + z/2)$, the checked identity is

$$z\mathcal{N}_{10}[h] = z^2P(z), \qquad P(z) = 189z^5 + 1778z^4 + 5492z^3 + 4848z^2 - 763z + 42.$$

The degree-five Bernstein check is recorded in Example B.1; its five positive minimum coefficients cover the entire interval $[-7/4, 1/4]$. As a separate exact check, let $P_0 = P$, $P_1 = P'$, and form the Sturm sequence by negative polynomial remainders. The endpoint sign vectors are

z	$(\operatorname{sgn} P_0(z), \ldots, \operatorname{sgn} P_5(z))$	sign changes
$-7/4$	$(+, +, -, -, +, +)$	2
$1/4$	$(+, +, +, -, -, +)$	2

Thus P has no real root in this interval; $P(0) = 42 > 0$ confirms its positive sign throughout.

S.6. Determinant recurrence and scalar inequalities. There are two kinds of checks in this group: a symbolic polynomial identity, and exact rational evaluations over specified finite ranges.

For $d = u + 1$, $r = v + 2d + 1$, put

$$B_{r,d} = \frac{2(r - d + 1)(r + 2d^2 + 3d + 2)}{(d + 1)(d + 2)} + 2d + 1, \qquad D_{r,d} = (r - 1)(d + 1)(d + 2)(r - d)^2.$$

Define

$$\Theta_{r,d} = \frac{2r^3}{r - 1}\left(\frac{1}{d + 1} - \frac{(r - 2d)(r - 2d - 1)}{(d + 2)(r - d)^2}\right) - B_{r,d}.$$

The all-variable identity checked by the program is

$$\begin{aligned} D_{r,d}\Theta_{r,d} = {} & 2uv^4 + (4u^3 + 25u^2 + 29u)v^3 \\ & + (16u^4 + 102u^3 + 206u^2 + 140u + 14)v^2 \\ & + (18u^5 + 137u^4 + 385u^3 + 484u^2 + 258u + 44)v \\ & + 4u^6 + 38u^5 + 140u^4 + 252u^3 + 230u^2 + 104u + 24. \end{aligned}$$

The numerator has nonnegative coefficients and positive constant term. Since $D_{r,d} > 0$ for $u, v \geq 0$, this identity supplies the sign used in the paper's telescoping estimate.

For the remaining checks, let e_j be elementary symmetric polynomials, with $e_0 = 1$. In m positive variables y_i, write

$$a_{si} = \frac{e_s(y_1, \ldots, \widehat{y_i}, \ldots, y_m)}{e_s(y)}, \quad p_i = \sum_{s=1}^{m-1} a_{si}, \quad D_m = \prod_{s=1}^{m-1} e_s,$$

and set

$$A=\sum_i y_i\left(\sum_{s=1}^{m-1} s\,a_{si}-\sum_{s=1}^{m-1} a_{si}^2\right),\qquad V=\sum_i y_i p_i^2,\qquad Y=\sum_i y_i^2 p_i,$$
$$\mathcal{C}_m=e_1D_m^2\left(2A+Y/e_1-V-2e_2/e_1\right).$$

The direct expression in the variables y_i is compared with its formal expression in $e_1,\dots,e_m$, obtained from

$$H_{ij}=\sum_{b=0}^{\min(i,j)}(i+j+1-2b)e_{i+j+1-b}e_b$$

and $H_{s0}^{(2)}=e_1e_{s+1}-(s+2)e_{s+2}$. Indices greater than m are set to zero in the formal expression defining $\mathcal{C}_m$.

The recurrence tested is

$$\mathcal{C}_{r+1}=e_r^2\mathcal{C}_r+\mathcal{R}_r,\qquad \mathcal{P}_r=e_1^3e_re_{r+1}\prod_{j=2}^{r-1}e_j^2,$$

where

$$\frac{\mathcal{R}_r}{\mathcal{P}_r}=r(r+1)-\sum_{p=1}^{\lfloor r/2\rfloor}(r+1)(2+\mathbf{1}_{p=1}+\mathbf{1}_{2p=r})\frac{e_r}{e_pe_{r-p}}$$
$$-\sum_{d=1}^{\lfloor (r-1)/2\rfloor}\sum_{j=1}^{r-2d}(r-j+1)(2+\mathbf{1}_{j=r-2d})\frac{e_je_r}{e_{j+d}e_{r-d}}.$$

For each $m=2,\dots,10$, the program evaluates the direct and formal expressions at $y=(1,2,\dots,m)$, checks their equality, and checks $\mathcal{C}_m(y)\ge 0$. For each $m=3,\dots,10$, it also checks the recurrence with $r=m-1$ at $e_j=e_j(1,\dots,N)$, for both $N=m$ and $N=m+2$. In the latter evaluation, the fixed formal polynomials $\mathcal{C}_m$ and $\mathcal{C}_{m-1}$ retain their respective definitions; their summation ranges are not enlarged to N.

Finally, define the scalar sums

$$I_r=\sum_{p=1}^{\lfloor r/2\rfloor}\frac{(r+1)(2+\mathbf{1}_{p=1}+\mathbf{1}_{2p=r})}{\binom{r}{p}},$$
$$J_r=\sum_{d=1}^{\lfloor (r-1)/2\rfloor}\sum_{j=1}^{r-2d}(r-j+1)(2+\mathbf{1}_{j=r-2d})\frac{\binom{j+d}{d}}{\binom{r}{d}}.$$

Exact rational arithmetic checks $I_r+J_r\le r(r+1)$ for every integer $r=2,\dots,100$. For example,

$$(I_2,I_3,I_4)=\left(6,4,\frac{25}{4}\right),\qquad (J_2,J_3,J_4)=\left(0,6,\frac{43}{4}\right).$$

The evaluations for orders 2 through 10, and for scalar indices 2 through 100, are finite consistency checks. The statements for arbitrary matrix order and all positive variables follow from the symbolic recurrence, Newton's inequalities, and the scalar estimates proved in the paper.

S.7. Hopf pairing and the intrinsic certificate. The seventh group applies the closure table (6.28) to check the quartic minimality equation, the three ambient potential identities (6.31)–(6.32), and the Jacobi weight $16R^2/T-8$. It also checks

$$|\nabla(4F-S^2)|^2=16R^3,\qquad \Delta(4F-S^2)=-8R.$$

For the intrinsic calculation it uses the ordered variables (R, F), the gradient matrix and drift

$$\begin{pmatrix} 4R & 8F \\ 8F & R^3 \end{pmatrix}, \qquad (62, 4F/R),$$

and verifies all three identities in (6.34). These ten checks are exact symbolic identities. The Clifford construction, geometric singular sets, and comparison across the vertex are proved in Section 6.3. In this group the code variable `r` denotes R, not $\sqrt{R}$.

S.8. Additional algebraic checks.

Clifford pullbacks (original group 8; Section 11). Starting with a harmonic Cartan–Münzner target of degree g, the program derives the source closure and checks

$$\det(\Gamma(R, Q, F)) = 256g^2R(R^2 - Q)(Q^g - F^2)$$

and both the abstract pullback defect and its expanded form for $G = QF$. It specializes to the cubic targets in Theorem 11.14. For unequal quadratic targets it checks the radial-power transfer formula, the two positive square completions in Example 11.12, and the source certificate. The boundary calculations give

$$\beta_{\mathrm{Cartan}} = (8m - 96)\frac{R^2}{Q} - 10, \qquad \beta_{\mathrm{Lawson}} = 24\frac{R^2}{Q} - 8, \qquad \left(8 - \frac{63}{2}\right)^2 + 16 = \frac{2273}{4}.$$

The group has 33 exact checks, including positivity of the relevant Cartan coefficients at $m = 14, 26$. In this group, the local code variables `t` and `q` denote the target squared radius $q = |y|^2$ and the source quantity Q, respectively.

Odd-degree rigidity and cubic spectrum (original group 9). The program checks the local second-order identities in Appendix D, including the maximum values, the Hessian eigenvalue relations, and the final residual

$$\frac{8(e-2)(e-3)}{e^2}.$$

It also checks the cubic spherical gradient and Laplacian table, the four moment identities before integration, and their elimination leading to $\alpha = 2(N + 3)/3$.

For the link calculation it checks the power-Jacobi identity, the smaller root, the cutoff exponent, and

$$9(k-1)^2 - B_k = (k + \delta_k)^2, \qquad H_5 = 42 + 10\sqrt{17}.$$

The quaternionic trace model is checked through its eigenvalue identities and Hessian trace, including the off-diagonal multiplicity four. The group distinguishes the bound 52 from $G = SF$ at $k = 5$ from the optimal value H_5. Together with the Simons and quartic identities, these are 32 exact checks. They verify the algebra used in the proofs, while the local singular-set geometry and Friedrichs-domain arguments remain analytic.

Critical correction and Lin's cone (original group 10; Chapter 5). At $d = 4g - 2$, $\ell = 2g$, and $h = z(1 + \lambda z)$, the coefficients of z^0, z^1, z^2 in $z\mathfrak{N}$ vanish and

$$[z^3](z\mathfrak{N}) = 2g^3((2g + 1)(1 - t^2)\lambda - 2gt).$$

The remaining three checks give $\Delta(RF)$, $|\nabla(RF)|^2$, and $\mathcal{L}(RF)$ for Lin's $\mathbb{R}^2 \times \mathbb{R}^6$ cone. There are seven checks in this group.

Changes of generators (original group 11; Chapters 5 and 6). Nine checks verify the six independent gradient entries and three Laplacians after changing the Hopf generators to $(R, S, Q) = (u + v, u - v, u^2 + v^2 - 2F)$. Four further checks verify the Lawson factorization (1.1) and the inverse expressions for the two squared block radii. All thirteen are exact polynomial or rational identities.

General multiplier identities (original group 12). Seven checks cover the two-jet expansion for an arbitrary multiplier, the scalar composition identity, the spherical numerator obtained directly in polar coordinates, the combined-power gradient and defect, and the two completed-square identities for the Hardy degree term and its change under radial perturbation. The combined-power calculation uses formal gradient contractions before any specialized closure is imposed.

S.9. Bihomogeneous quartics and the singular-link obstruction. The last two original groups check the closure calculations of Section 6, Proposition 11.8, and Theorem 11.11. For the splitting quartic, the input table is

```
vars = {u,v,q}; rr = u+v; tt = u v;
matrix = {{4u,0,4q},{0,4v,4q},{4q,4q,4rr q}};
drift = {2 ell,2 ell,2 mm rr};
f = q-c tt; a = 1-2c; b = c(1-c);
minimalRule = mm -> 1+c(ell-2);
```

The exact checks compare the degree-eight and degree-ten divergence numerators with (6.15) and (6.20), verify the endpoint and discriminant formulas used for positivity, and recover (6.22) through the Bochner identity. The higher-degree checks substitute the minimal levels into the octonionic pullback closure.

For Riedler's $C^4_{16,4}$, the Euclidean closure independently gives (11.21). Its angular calculation is

```
phi = 1/q;
Factor[16 q(1-q) D[phi,{q,2}]
  +(56-64q) D[phi,q]+(24/q+14)phi-46phi]
(* 0 *)
(13/2)^2-46
(* -15/4 *)
```

These outputs verify the displayed algebraic identities. The cutoff estimate $O(\varepsilon)$ in Theorem 11.11 is needed to turn the formal angular function into admissible variations. The final group also checks the general orthogonal-quartic numerator and the Hopf incidence cubic table.

S.10. Clifford identities and rational Jacobi moments. The stable unit `Chapter08` contains 193 assertions. Its first three blocks preserve the earlier power-multiplier and Bernstein certificates; the present minimizing proof uses Proposition 9.5, whose transport and parameter identities are checked in the separate cubic-update unit. The remaining blocks check the Jacobi matrix, explicit quaternionic Clifford systems, and the Hopf norm and spherical-gradient Gram identities. The coordinate contractions are tested at five exact points for each of $(m,q)=(4,3),(4,4)$; these finite tests supplement the formal derivation in `notes/Clifford_Hessian_Identities.pdf`. The Hopf identities are checked as polynomial identities in eight coordinate variables.

For the Jacobi calculation in (9.30), set

$$U_i = \mathrm{B}\left(i+3,\frac{3}{2}\right), \qquad V_j = \frac{1}{\pi}\mathrm{B}\left(j+\frac{3}{2},\frac{1}{2}\right).$$

The beta-function recurrence gives

$$U_0 = \frac{16}{105}, \qquad U_i = \frac{2(i+2)}{2i+7}U_{i-1} \quad (i\ge 1),$$
$$V_0 = \frac{1}{2}, \qquad V_j = \frac{2j+1}{2j+2}V_{j-1} \quad (j\ge 1).$$

A monomial $u^i\theta^j$ in the polynomial part of the integrand therefore contributes U_iV_j after division by π. The direct Gamma-function calculation and these rational recurrences both give

$$\mathsf{M}_J = \begin{pmatrix} 96/715 & 471936/1616615 \\ 471936/1616615 & 571904/969969 \end{pmatrix}, \qquad (1,-1/2)\mathsf{M}_J(1,-1/2)^T = -\frac{9952}{969969},$$

recovering (9.31). Here M_J is the Jacobi matrix; the scalar contractions remain $J = |\nabla S|^2$ and $K = \Delta S$.

S.11. Verification units and their scope. The retained units have filenames of the form `Verify_Chapter_04.wl` or `Verify_Appendix_A.wl`, with matching notebooks in `mathematica/chapters`. Additional units are in `mathematica/supplemental`; their full paths and commands are listed in `README.txt`. The counts below are program assertions, including stated repetitions, rather than distinct mathematical results.

Unit	Current material	Checks
Chapter04	4: algebraic reduction and Jacobi stability	6
Chapter05	5: isoparametric hypercones	16
Chapter06	6: bihomogeneous hypercones	49
Chapter07	7.1, 8.1: gradient-square closure and cubics	18
Chapter08	9: Clifford cubic cones	193
Chapter09	11: retained pullback examples	50
Chapter10	12: complex determinants	134
AppendixA	A: power multipliers	8
AppendixB	B: positivity certificates	15
AppendixC	C: singular cutoff exponent	1
AppendixD	D: odd-degree rigidity and moments	15
AppendixE	Ancillary Clifford Hessian contractions	18
AppendixF	Historical homogeneity and limiting-parameter checks	27
Retained suite	Thirteen units	550
General cubic	7: multiplication dictionary and SF identities	31
Exceptional cubics	Jordan and G_2 four-generator identities	28
Cubic update	Common potential, Clifford and nonradial certificates, nonharmonic Appendix F	38
Cartan pencil	One-parameter identities and regular-zero obstruction	65
Intrinsic table	Real- and complex-coordinate checks of the five-generator table	62
Pullback transfer	Transfer identities, dimension monotonicity, and Simons target	13
All units		787

The general cubic unit allows a formal nonscalar trace $K = \Delta S$; the retained Chapter07 unit checks its specialization $K = 4kR$. The exceptional and cubic-update units check identities from their input differential tables. The Cartan-pencil unit also differentiates the real model directly; the intrinsic-table unit differentiates both the real and complex models, including all gradient products, Laplacians, recovery formulas, and a nonzero rank-five Jacobian minor. These model checks do not replace the general proof of Theorem 10.13.

The determinant unit includes the symbolic identity in S.6 and finite checks for orders up to ten and scalar indices up to one hundred; the statements in arbitrary order are proved in Section 12. Likewise, the cutoff-exponent check does not establish the singular-set geometry or the analytic cutoff construction. The historical `AppendixF` unit does not correspond to the present nonharmonic Appendix F, which is covered by the cubic-update unit.

S.12. The pullback transfer identities. The additional unit `mathematica/supplemental/pullback_transfer_20260930` checks the gradient, minimality-operator, and ratio identities used in Theorem 11.4, together with a specialization consistency check of the weighted Laplacian formula and the target calculation for $H = q^{3/4} f$ in Example 11.10. It also checks the finite difference in (11.13). Its thirteen exact checks passed in the supplied Python implementation using rational arithmetic. The accompanying Wolfram Language implementation also passed all thirteen checks in an author-run native Mathematica 12.0.0 execution on 2 October 2026. The evaluated notebook and TXT, CSV, and JSON records are supplied in the same directory. The weighted Laplacian check specializes the stated formula; independent second-order differentiation is performed in the retained verification unit `AppendixA`. These identities support the differential calculation; the smallest-algebra, singular-set, and iteration arguments are proved in the text.